

Hazara Division AQI Prediction System

Real-Time Environmental Monitoring & Forecasting

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10Pearls Internship Project

June 2026

Project Overview

- **Objective:** Provide localized hourly tracking and 72-hour air quality forecasts for Hazara Division, KPK.
- **Scope:** Covers 6 key districts: Abbottabad, Haripur, Mansehra, Battagram, Upper Kohistan, and Torgar.
- **Core System:**
 - Automated ingestion of weather and air pollutant concentrations.
 - 6 predictive models (Linear, Ridge, RF, Gradient Boosting, XGBoost, LSTM).
 - Interactive FastAPI dashboard for real-time visualization and data exports.

The Valley Inversion Phenomenon

Unlike open coastal regions, Hazara Division's geography features deep mountain valleys that trap emissions.

- **Temperature Inversions:** Cold air settles in valley basins overnight, trapping pollutants close to the ground.
- **Topographical Barriers:** High mountains block horizontal wind dispersion, creating persistent local pollution.
- **Seasonal Drivers:** Winter wood heating, dust, and agricultural waste burning cause sharp spikes in particulate matter.

Data Pipeline & Ingestion

- **Data Source:** Open-Meteo Air Quality API.
- **Target Metric:** US Air Quality Index (US AQI).
- **Pollutants Monitored:**
 - Particulate Matter (PM2.5, PM10).
 - Gaseous species (CO, NO2, SO2, Ozone).
- **Data Cleaning:**
 - Handled extreme outliers by clipping at the 1st and 99th percentiles.
 - Filled missing intervals using forward-fill (FFill) and backward-fill (BFill).

We engineer 13 dynamic features to capture persistence and trends:

- **Lag Features:** 1-hour, 2-hour, and 24-hour lags of AQI to model short-term persistence.
- **Rolling Statistics:** 6-hour and 24-hour moving averages for PM2.5 and overall AQI to track background build-up.
- **Rate of Change:** First-order difference of the AQI value to capture build-up velocity.
- **Diurnal & Seasonal Indicators:** Time of day, day of the week, and month of the year.

Model Comparisons

Models evaluated using chronological 80/20 train/test split:

Model Type	Category	RMSE	MAE	R^2 Score
Linear Regression	Statistical	4.6796	2.2829	0.8659
Ridge Regression	Regularized	4.6797	2.2831	0.8659
XGBoost	Boosting	5.8714	3.0906	0.7891
Gradient Boosting	Boosting	5.9938	3.2084	0.7802
Random Forest	Ensemble	7.4568	3.8614	0.6593
LSTM	Neural Network	10.0665	7.9150	0.3781

- **Linear Performance:** Linear and Ridge regression models achieved the lowest error. Short-term air quality is highly linear (correlation coefficient of lag-1h with target AQI exceeds 0.95).
- **Deep Learning Representation:** LSTM models capture temporal dependencies over the past 24 hours. While standalone metrics are higher, it handles non-linear dynamics effectively.

Explainability (SHAP Values)

To verify predictions, we calculated global SHAP importance for the primary model:

- **1. Lag-1h AQI** (Most critical): Dictates immediate persistence (current air directly affects the next hour).
- **2. 24h Rolling AQI**: Represents background accumulations and slow dispersion patterns.
- **3. PM2.5 24h Average**: Confirms fine particulate matter as the major driving pollutant across all districts.
- **4. Diurnal Hour**: Captures morning/evening rush hour patterns and daily thermal cycles.

Dashboard & Web Interface

- **Backend:** FastAPI for high performance, request routing, and asynchronous API endpoints.
- **Frontend UI:** Clean dark theme built with vanilla HTML/CSS.
- **Key Features:**
 - District Selection: Quick switching across all six districts.
 - Model Selection: Interactive leaderboard showing active metrics.
 - Interactive Charts: Embedded Plotly visualizations for history and forecasts.
 - Explainability Panel: Live rendering of SHAP feature importance.
 - Data Export: Single-click download of full 72-hour forecast CSVs.

Production Automation (MLOps)

- **Local Controller:** 'run.sh' script to manage execution, status checks, and local manual pipelines.
- **Automated Ingestion:** GitHub Actions workflow runs every hour to fetch raw readings, clean data, and update CSV storages.
- **Automated Retraining:** Daily GitHub Actions workflow retrains all models, computes fresh validation metrics, regenerates SHAP plots, and pushes update files.
- **Automatic Redeployments:** Connected to Render hosting platform for instant deployments on commit.

Conclusion & Future Outlook

- Successfully built and deployed a production-grade AQI prediction platform.
- Solved key localization issues including valley inversion modeling and local timezone alignment.
- **Future Work:**
 - Integrate local physical meteorological models (WRF) for wind vectors.
 - Incorporate holiday and industrial activity schedules into features.
 - Expand coverage to additional districts in KPK.